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ShowName: Cisco CCNA (200-301)
Topic: 1.0 Networking Fundamentals
EpisodeName: IP Addressing: IPv6 Basics & Address Types Pt2

Description: In this episode, Aubri and Ronnie look into the growing use of IPv6. They discuss the different types of IPv6 addresses that candidates should be able to identify and understand the usage scenario of them.

IP Addressing: IPv6 Basics & Address Types

1.9 Compare IPv6 address types

<u>Basic</u>

- Compare with IPv4 (32bit vs 128bit)
- Analogies
- Format: Hexadecimal e.g. 2001:1fbc:2cca:3ddb:a123:b234:c345:d456/64

<u>1.9.a Global unicast</u>

- 2000::/3 - 3fff::/3
- Routable IPv6 address on the internet = IPv4 public address
- Devices can be configured

Manually configured
Stateless Address Autoconfiguration
Stateful DHCPv6

<u>1.9.b Unique local</u>

- FC00::/7 - FDFF::/7
- Equivalent to IPv4 private addresses

<u>1.9.c Link local</u>

- FE80::/10 - (FEBF::/10)
- Unique only on it's own subnet
- not forwarded by routers
- only one link-local address per interface
- Device Configuration

Can create own link-local address when device start
Can be manually configured

- Role:

Used to communicate with other devices on network
A router's link local address is the default gateway for other addresses
used to exchange IPv6 dynamic routing information (messages and routing information)

<u>1.9.d Anycast</u>

Can be assigned to to more than one device.
routed to the "nearest" device with that anycast address according to routing table of the router.
no special prefix but uses same as global unicast

<u>1.9.e Multicast</u>

- FF00::/8
- One to Many
- Well known multicasts:

FF02::1 — all IPv6 devices
FF02::2 — all IPv6 routers
FF02::5 — all OSPFv3 routers
FF02::A — all EIGRP (IPv6) routers

1.9.f Modified EUI 64

Refer to notes EUI-64